6. Use Figma to simulate a handoff scenario in a mobile network and visualize the process.

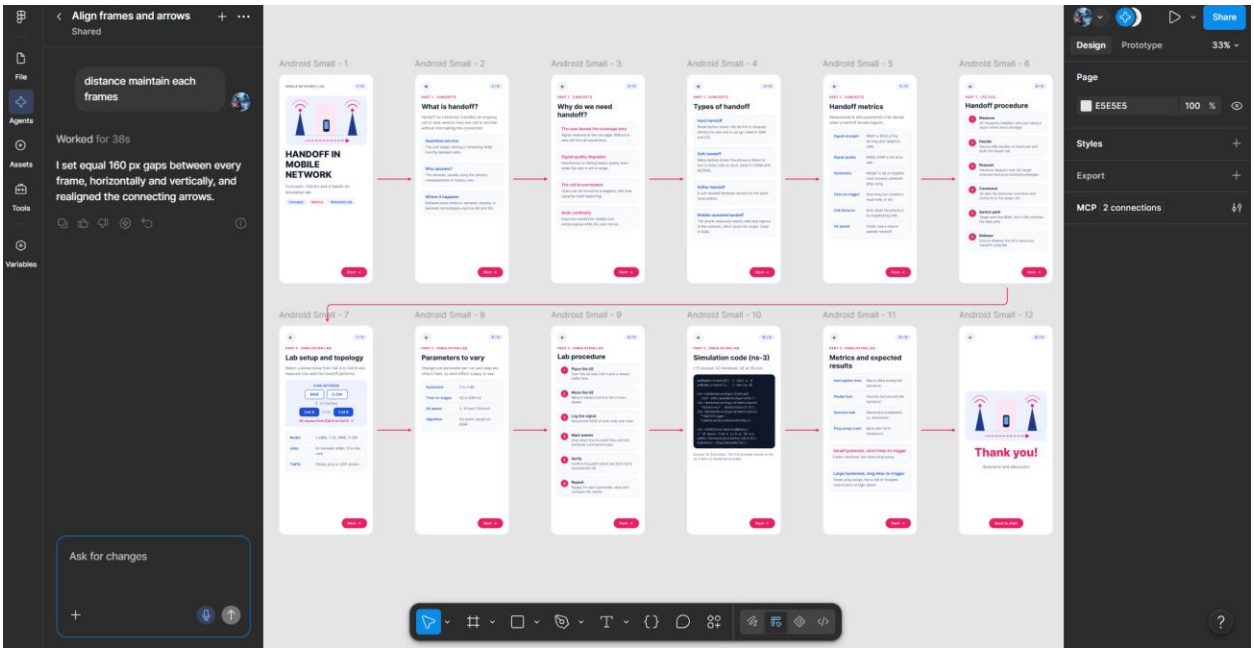
Aim: -

To simulate a handoff scenario in a mobile network and visualize the process using Figma.

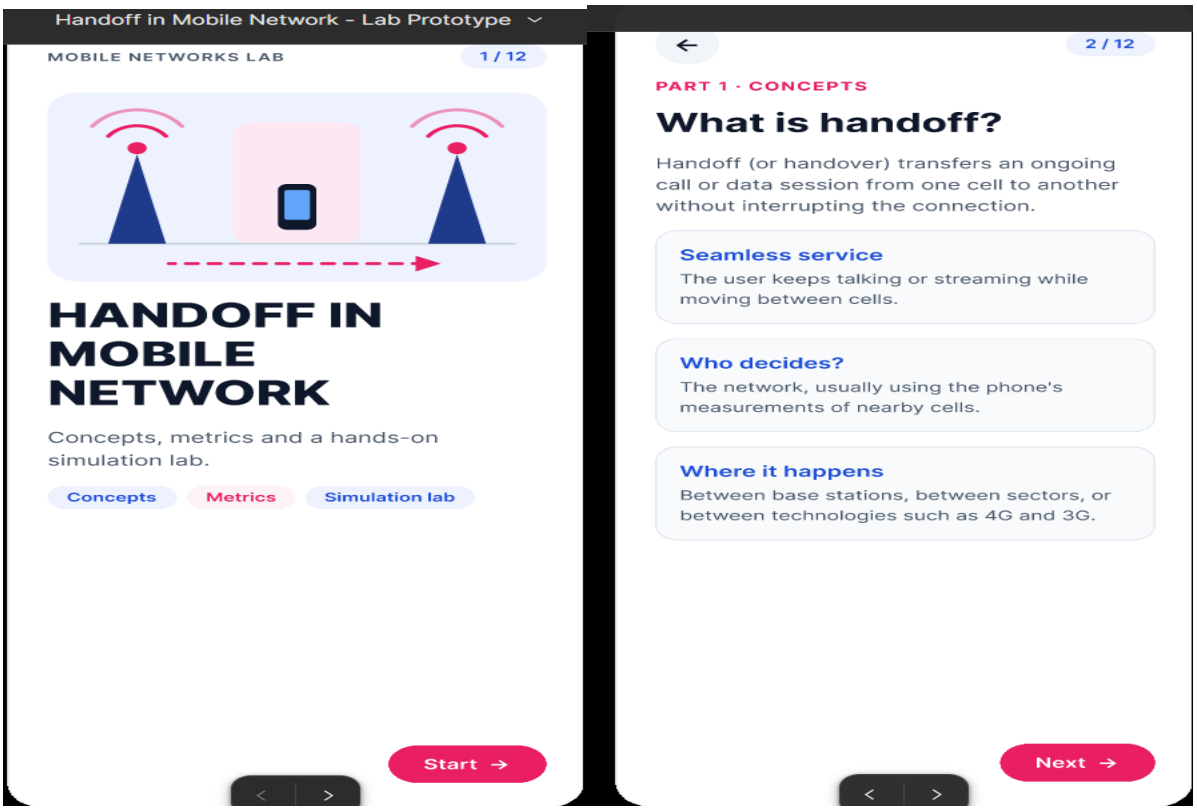
Procedure: -

1. open Figma
2. Create a new file
3. Select the Frames
4. Design Visual Elements
5. Make it Interactive
6. Add icons on the Frame
7. Incorporate Multimedia
8. Storyboard Animation
9. Review and edit the Prototype
10. Save and Share

Design :-


Align frames and arrows
Shared
distance maintain each frames
Worked for 38s
I set equal 160 px gaps between every frame, horizontally and vertically, and realigned the connecting arrows.
Ask for changes
Android Small - 1
HANDOFF IN MOBILE NETWORK
Android Small - 2
What is handoff?
Android Small - 3
Why do we need handoff?
Android Small - 4
Types of handoff
Android Small - 5
Handoff metrics
Android Small - 6
Handoff procedure
Android Small - 7
Lab setup and topology
Android Small - 8
Parameters to vary
Android Small - 9
Lab procedure
Android Small - 10
Simulation code (ns-3)
Android Small - 11
Metrics and expected results
Android Small - 12
Thank you!
Share
Design
Prototype
33%
Page
E5E5E5
100 %
Styles
Export
MCP 2 connections

Prototype:-


Handoff in Mobile Network - Lab Prototype
MOBILE NETWORKS LAB
1 / 12
HANDOFF IN MOBILE NETWORK
Concepts, metrics and a hands-on simulation lab.
Concepts
Metrics
Simulation lab
Start →
2 / 12
PART 1 · CONCEPTS
What is handoff?
Handoff (or handover) transfers an ongoing call or data session from one cell to another without interrupting the connection.
Seamless service
The user keeps talking or streaming while moving between cells.
Who decides?
The network, usually using the phone's measurements of nearby cells.
Where it happens
Between base stations, between sectors, or between technologies such as 4G and 3G.
Next →

3 / 12

PART 1 · CONCEPTS

Why do we need handoff?

The user leaves the coverage area

Signal weakens at the cell edge. Without a new cell the call would drop.

Signal quality degrades

Interference or fading lowers quality even while the user is still in range.

The cell is overloaded

Users can be moved to a neighbor with free capacity (load balancing).

Goal: continuity

Keep the connection reliable and uninterrupted while the user moves.

Next →

4 / 12

PART 1 · CONCEPTS

Types of handoff

Hard handoff

Break-before-make: the old link is released before the new one is set up. Used in GSM and LTE.

Soft handoff

Make-before-break: the phone is linked to two or more cells at once. Used in CDMA and WCDMA.

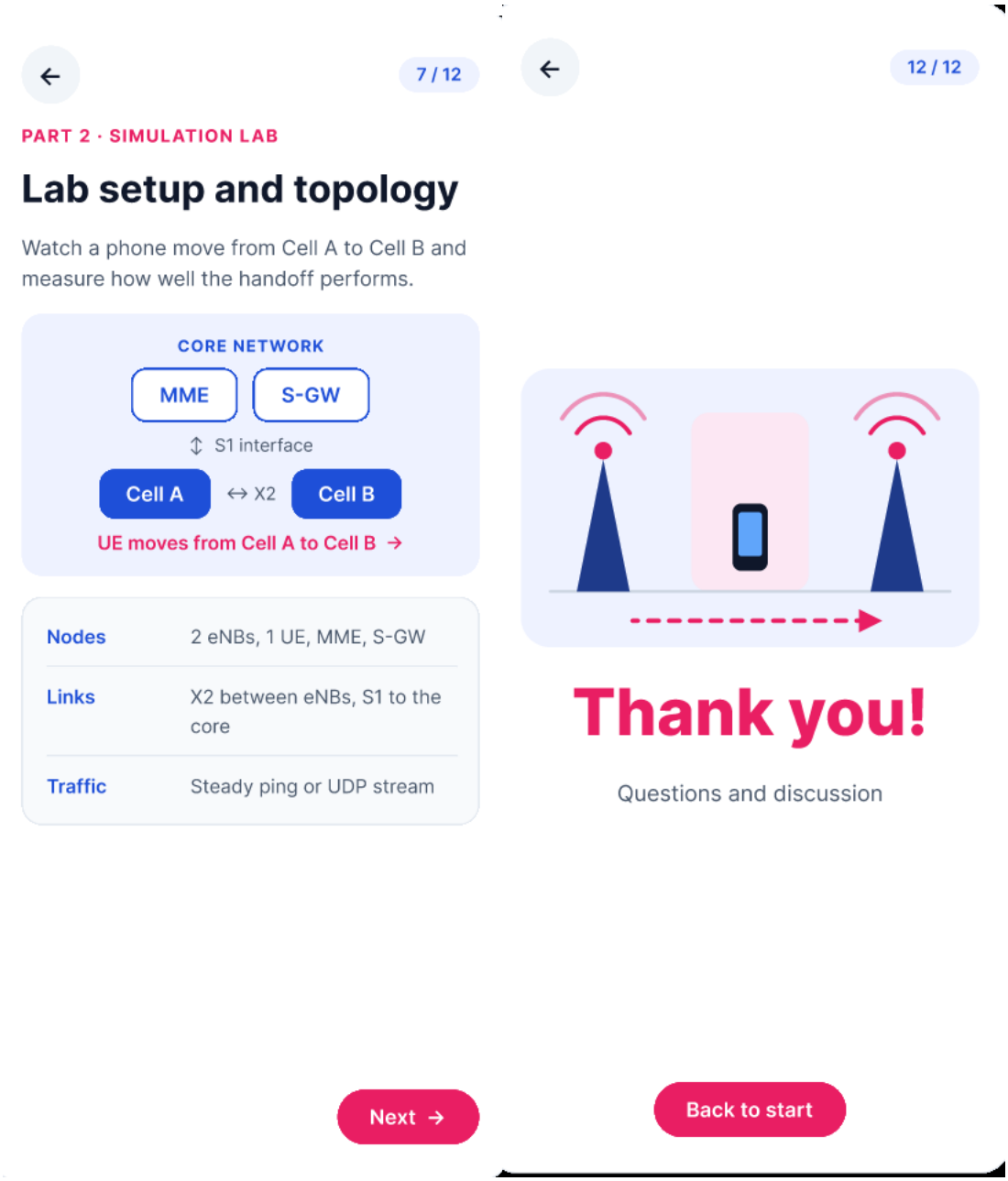
Softer handoff

A soft handoff between sectors of the same base station.

Mobile-assisted handoff

The phone measures nearby cells and reports to the network, which picks the target. Used in GSM.

Next →



Result: -

Hence the simulated a handoff scenario in a mobile network and visualized the process using Figma.